

The Ironclad Credentials Vault

AES-256-GCM Encryption: Zero Secret Visibility in Model Prompts

Protecting Secrets Against Prompt Injection & Extraction

A major vulnerability in AI wrappers is passing API keys inside prompt contexts. **Fathom isolates credentials behind an AES-256-GCM encrypted vault.**

Fathom			RING CRATE · HARDWARE KEY DERIVATION
KEY NAME	CIPHERTEXT DIGEST	PROMPT EXPOSURE	
hubspot_api_key	\$aes-gcm\$vl\$e8f9a2b01c4...	0% (NEVER PASSED TO LLM)	
amocrm_oauth_token	\$aes-gcm\$vl\$991a0f8b2c1...	0% (NEVER PASSED TO LLM)	

1. HARDWARE-GRADE ENCRYPTION (RING CRATE)

- Secrets stored in SQLite are encrypted using **AES-256-GCM** authenticated encryption.
- Encryption keys are derived from secure environment variables (FATHOM_CREDENTIAL_KEY).
- API listing endpoints return strictly masked strings (e.g. sk-live-***1234).

2. ZERO SECRET TOOLS IN AGENT REGISTRY

- Agents have **zero tools to read or query plaintext secrets**.
- When an agent invokes a service, the backend adapter resolves credentials internally in Rust memory.
- Prompt injection attacks cannot extract keys that the LLM has no mechanism to access.

Enterprise Compliance: Enterprise security teams can deploy Fathom with full confidence that proprietary CRM tokens and database passwords remain completely secure.